A CHILDCARE MANAGEMENT SYSTEM FOR SERVICE COORDINATION AND REPORTING IN MOMBASA

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DECLARATION

I declare that this project is my original work and has not been presented for a degree in any other University or for any other award.

Student's Name: Grace Mucheru Waithira
Sign: _____

Date: _____

I confirm that the work reported in this project was carried out by the candidate under my supervision.

Supervisor's Name: Dr. Kennedy Hadullo
Sign: _____

Date: _____

DEDICATION

I dedicate this project to parents, guardians, and caregivers in Mombasa who strive daily to balance work and childcare responsibilities. Their challenges inspired the need for a structured, digital solution to improve service delivery and accountability.

ACKNOWLEDGEMENT

I sincerely thank my supervisor for his guidance and constructive feedback throughout this project. I also appreciate the childcare centers in Mombasa that provided insights into their operational challenges, which shaped the requirements of this system.

ABSTRACT

Childcare services in Mombasa face challenges in managing enrollment, attendance, billing, and communication between parents and caregivers. Manual record-keeping often leads to inefficiencies, errors, and lack of transparency. This project proposes the design and development of a **Childcare Management System (CMS)** , a ReactJS-based web application supported by Yarn and Bootstrap CDN for responsive design. The system will integrate modules for child enrollment, attendance tracking, staff scheduling, billing, and parent communication. Data collection will involve interviews and questionnaires with childcare providers and parents in Mombasa. The system will be evaluated based on usability, functionality, and performance. Findings are expected to show that a modular, web-based approach improves efficiency, accountability, and parent satisfaction compared to manual methods. Future enhancements may include mobile app integration and AI-driven analytics for child development tracking.

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List of Abbreviations & Acronyms

- **CMS**: Childcare Management System
- **UI**: User Interface
- **UX**: User Experience
- **CRUD**: Create, Read, Update, Delete
- **CSV**: Comma-Separated Values
- **JSON**: JavaScript Object Notation
- **DB**: Database
- **SQL**: Structured Query Language
- **HTTP**: Hypertext Transfer Protocol
- **HTTPS**: Hypertext Transfer Protocol Secure
- **REST**: Representational State Transfer
- **SDLC**: Software Development Life Cycle
- **UML**: Unified Modeling Language
- **IDE**: Integrated Development Environment
- **CLI**: Command Line Interface

- **CDN**: Content Delivery Network
- **SSL**: Secure Sockets Layer
- **ORM**: Object-Relational Mapping
- **MVC**: Model-View-Controller

1.0 CHAPTER ONE: INTRODUCTION

1.1 Introduction

Childcare services are essential for supporting working families and promoting early childhood development. According to UNESCO (2021), early childhood care and education (ECCE) is a critical foundation for lifelong learning, social well-being, and national development. Quality childcare not only provides a safe environment for children but also enables parents to participate productively in the workforce. Despite this importance, many childcare centers in Kenya still rely on fragmented manual processes such as paper-based enrollment, handwritten attendance registers, and informal communication channels. These outdated practices hinder efficiency, reduce accountability, and limit the ability of administrators to make data-driven decisions. With the rapid growth of digital technologies, there is an urgent need to modernize childcare management systems. A centralized Childcare Management System (CMS) can streamline operations, improve communication between parents and caregivers, and enhance service quality. By leveraging web-based platforms, childcare centers can achieve transparency, scalability, and improved trust among stakeholders.

1.2 Background of the Study

In Mombasa, childcare centers face significant operational challenges. Manual record-keeping often results in misplaced files, duplication of data, and delays in accessing critical information. Communication with parents is inconsistent, typically relying on phone calls or informal notes, which can lead to misunderstandings and reduced parental engagement. Billing systems are frequently inefficient, with errors in fee calculation and delays in issuing receipts, undermining financial accountability.

Research shows that digital management systems improve transparency, efficiency, and service delivery in educational and childcare institutions (Abu Yazid et al., 2025). Globally, platforms such as Brightwheel and HiMama have demonstrated the benefits of integrating enrollment, attendance, billing, and communication into unified systems. However, these solutions are often costly and not tailored to the realities of small and medium childcare centers in Kenya. A web-based system designed for local needs can consolidate these

functions into one accessible platform, ensuring affordability, ease of use, and adaptability to the Mombasa context.

1.3 Problem Statement

Current childcare management practices in Mombasa are inefficient, relying heavily on paper records and ad-hoc communication methods. This fragmented approach leads to:

- Errors in billing and fee collection.
- Poor attendance tracking, making it difficult to monitor child participation.
- Limited visibility for parents regarding their child's daily activities and progress.
- Administrative difficulties in managing staff schedules and workloads.

Without a centralized system, administrators struggle to maintain accountability, parents lack timely updates, and staff face unnecessary workloads. These inefficiencies compromise service quality, reduce parental trust, and hinder the scalability of childcare centers.

1.4 Purpose of the Study

The purpose of this study is to design and develop a modular Childcare Management System tailored to the needs of childcare centers in Mombasa. The system aims to enhance visibility, accountability, and communication by consolidating enrollment, attendance, billing, and parent-caregiver interaction into a single web-based platform.

1.5 Specific Objectives

i. To analyze existing childcare management practices in Mombasa and identify user requirements.

ii. To design a modular web-based system integrating enrollment, attendance, billing, and communication.

iii. To develop the system using ReactJS, Yarn, and Bootstrap CDN for responsiveness and scalability.

iv. To test the system for usability, accuracy, and performance in real-world childcare environments.

1.6 Justification of the Study

The proposed CMS will reduce manual effort, minimize errors, and improve efficiency in childcare service delivery. Parents will benefit from real-time updates on their children's attendance and activities, while administrators will gain structured reports for decision-making and financial accountability.

From an academic perspective, the project applies theoretical knowledge of system design, software engineering, and information management to a practical, community-based problem. It also contributes to the growing body of research on digital transformation in educational and childcare institutions, demonstrating how open-source technologies can be leveraged to create affordable, scalable solutions in developing contexts (Pressman, 2018; Sommerville, 2016).

1.7 Expected Challenges

The development and deployment of the CMS may encounter several challenges:

- **Incomplete Requirements:** Stakeholders may not fully articulate their needs during initial consultations.
- **Resistance to Adoption:** Some administrators and caregivers may prefer traditional methods due to familiarity.
- **Technical Issues:** Internet connectivity and device limitations may affect system accessibility.
- **Resource Constraints:** Limited budgets may restrict hardware acquisition or training opportunities.

These challenges will be mitigated through iterative development, stakeholder consultation, modular design, and user training. Agile Scrum methodology will ensure flexibility and continuous improvement throughout the project lifecycle.

2.0 CHAPTER TWO: LITERATURE REVIEW

2.1 Introduction

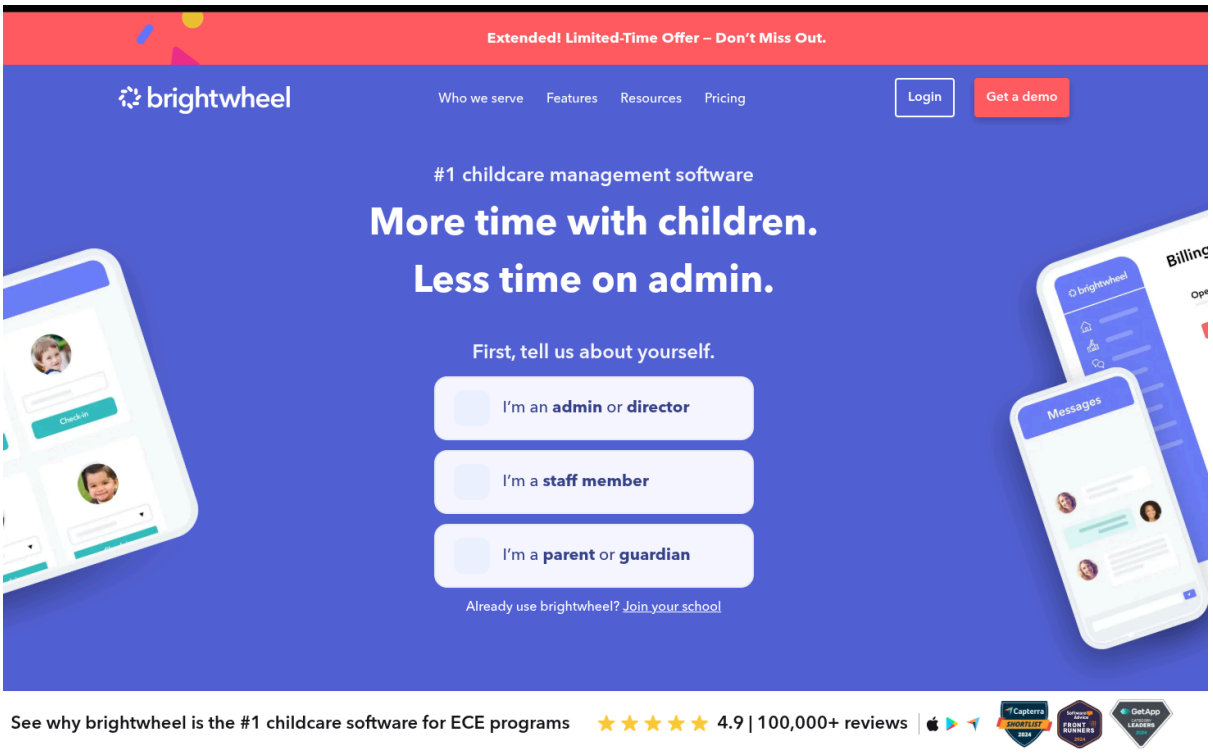
Childcare management systems are increasingly recognized as essential tools for improving efficiency, transparency, and accountability in early childhood care institutions. According to UNESCO (2021), digital solutions in education and childcare enhance service delivery by reducing administrative burdens and improving communication between stakeholders. In Kenya, most childcare centers still rely on manual processes, which are prone to errors and inefficiencies. Reviewing existing systems provides insight into their strengths and weaknesses, and highlights the research gap that this project seeks to address.

2.2 Review of Existing Similar Systems

2.2.1 Brightwheel

Brightwheel is a widely adopted childcare management platform in North America. It offers features such as digital check-in/check-out, parent communication, and billing integration.

URL: https://mybrightwheel.com/

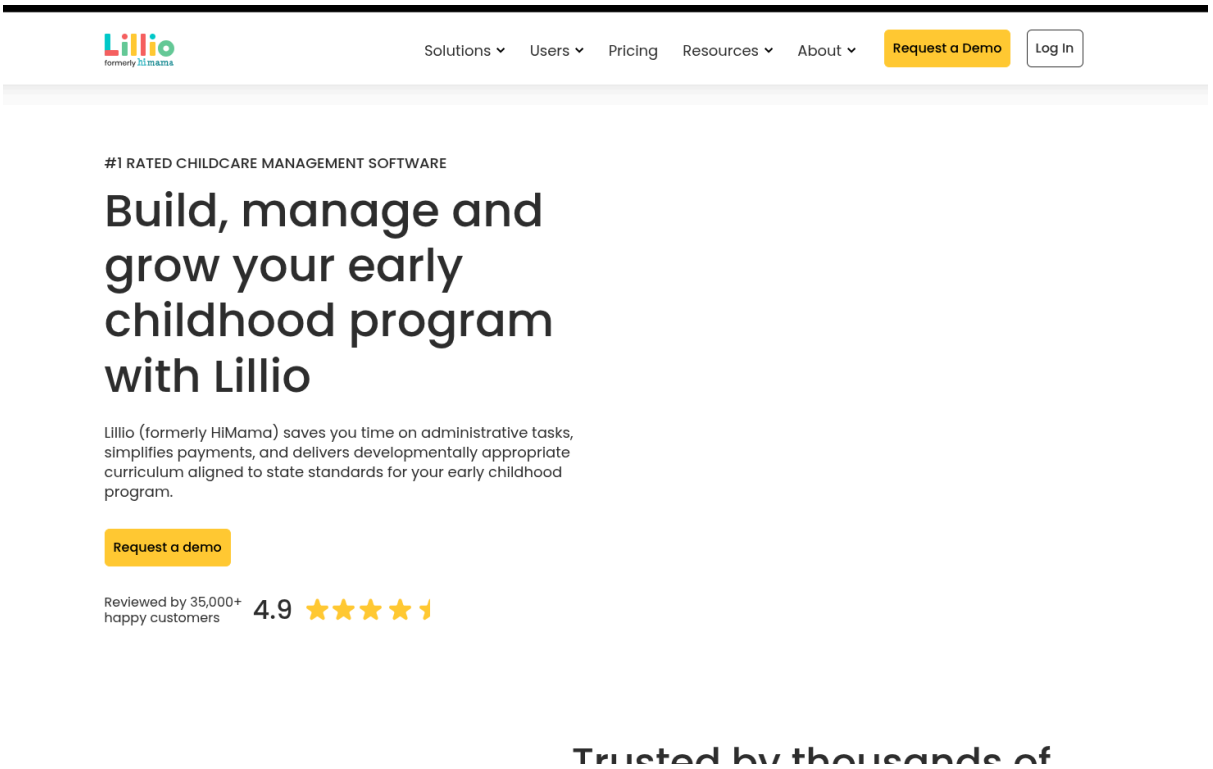


- **Strengths:** Comprehensive functionality, strong parent engagement tools, mobile app support.
- **Weaknesses:** Subscription costs are high, making it inaccessible for small centers in developing regions.

2.2.2 HiMama

HiMama focuses on parent communication and developmental tracking. It allows teachers to share daily reports, photos, and developmental milestones with parents.

URL: https://www.lillio.com/



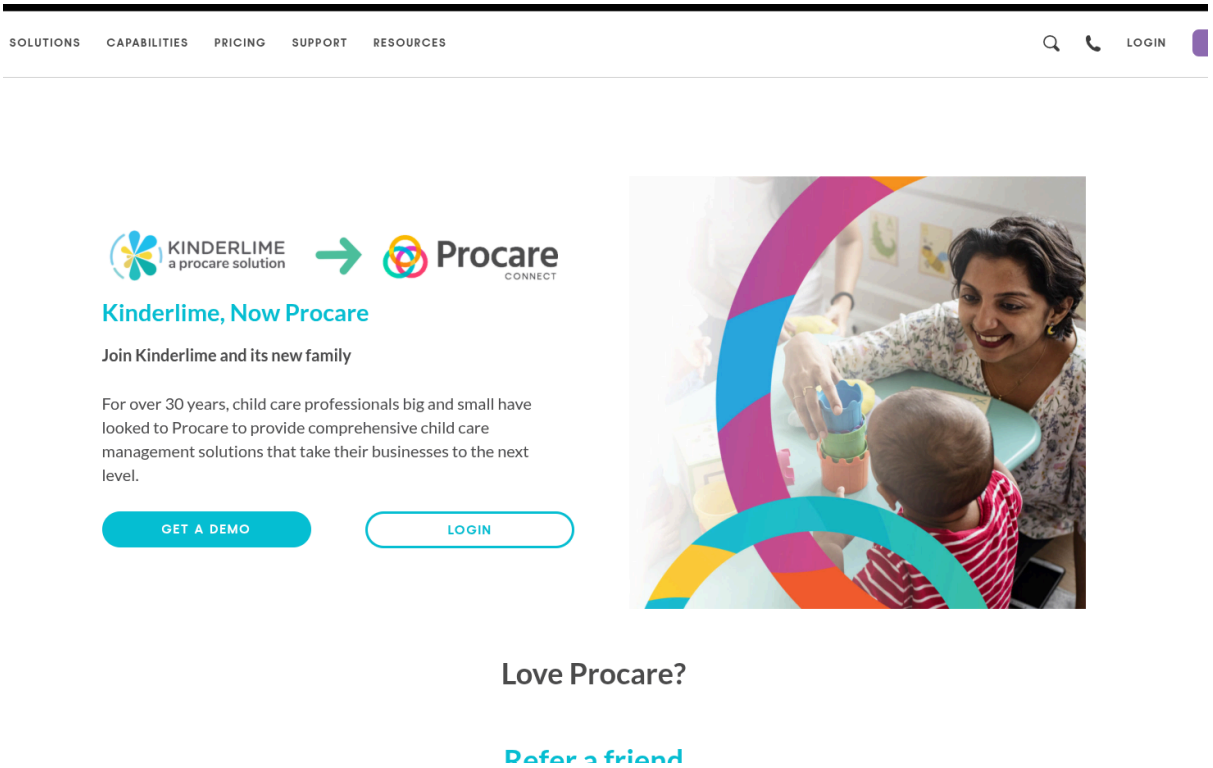
- **Strengths:** Rich communication features, strong focus on child development reporting.
- **Weaknesses:** Proprietary system, limited customization, and requires ongoing subscription fees.

2.2.3 Kinderlime

Kinderlime emphasizes billing and attendance management. It provides digital timesheets, staff scheduling, and invoicing tools.

URL:

https://www.procaresoftware.com/kinderlime-is-part-of-the-procare-family-of-brands/



- **Strengths:** Simplifies billing and payroll, integrates attendance tracking.
- **Weaknesses:** Limited scalability, lacks modern web frameworks, and minimal focus on parent communication.

2.4 How the Childcare Management System (CMS) is Better

The proposed CMS improves upon existing systems in several key ways:

1. **Affordability**
    - Built on open-source technologies (ReactJS, Yarn, Bootstrap CDN).
    - Eliminates costly subscription fees associated with commercial platforms.
2. **Localization**
    - Tailored to the operational realities of childcare centers in Mombasa.
    - Supports local languages and cultural practices in communication modules.
3. **Integration**
    - Combines enrollment, attendance, billing, and communication into one unified platform.

    - Reduces reliance on fragmented tools and manual record-keeping.
4. **User-Friendly Design**
    - Responsive interface optimized for mobile devices.
    - Simple dashboards for administrators, staff, and parents.
5. **Transparency & Accountability**
    - Structured reports in CSV/JSON formats for easy sharing and auditing.
    - Timestamped logs for attendance and billing records.
6. **Scalability**
    - Modular architecture allows new features (e.g., developmental tracking, analytics) to be added easily.
    - Suitable for both small centers and larger institutions.
7. **Educational Value**
    - Provides students and IT staff with practical exposure to modern web-based systems.
    - Serves as a teaching tool for demonstrating system design and software engineering principles.
8. **Security & Privacy**
    - Incorporates HTTPS, SSL, and secure authentication protocols.
    - Ensures sensitive child and parent data is protected.

2.4 Summary and Research Gap

While Brightwheel, HiMama, and Kinderlime provide valuable features, none fully satisfies the combined needs of affordability, localization, and usability for childcare centers in Mombasa. Existing systems are either too costly, too complex, or insufficiently adaptable to local contexts. The proposed Childcare Management System (CMS) addresses this gap by offering:

- A modular design integrating multiple functions (enrollment, attendance, billing, communication).
- A responsive, mobile-friendly interface accessible even on low-cost devices.
- Structured reporting for administrators and parents.
- Lightweight deployment using open-source technologies, reducing costs.

2.5 Literature Review on System Design and Development

System analysis involves identifying user needs and childcare requirements. According to Pressman (2018), effective analysis reduces risk of failure by aligning tools with organizational goals. System design defines architecture, interfaces, and reporting formats, emphasizing modularity and usability (Sommerville, 2016). Implementation translates design into functional modules, while testing ensures accuracy and usability.

3.0 CHAPTER THREE: METHODOLOGY

3.1 Introduction

This study adopts the **Software Development Life Cycle (SDLC)** methodology, guided by **Agile Scrum** for iterative development. SDLC provides a structured framework for system development, ensuring that analysis, design, implementation, and testing are systematically addressed (Pressman, 2018). Agile Scrum complements this by introducing flexibility, stakeholder involvement, and continuous feedback, which are particularly important in dynamic environments such as childcare centers where requirements may evolve over time. By combining SDLC and Agile Scrum, the project ensures both rigor and adaptability, allowing the Childcare Management System (CMS) to be developed in phases while incorporating user feedback at each stage.

3.2 Data Collection Methods

To ensure the CMS reflects actual user needs in Mombasa childcare centers, multiple data collection methods will be employed. This triangulation enhances the reliability and validity of findings (Creswell, 2014).

- **Interviews:** Semi-structured interviews with childcare administrators and parents will be conducted to identify pain points in current practices. These interviews will provide qualitative insights into challenges such as billing errors, poor communication, and attendance tracking gaps.
- **Observation:** Direct observation of childcare workflows will help capture real-time practices, including how staff record attendance, communicate with parents, and manage schedules. Observation ensures that the system design reflects actual operational realities rather than assumptions.
- **Questionnaires:** Structured questionnaires will be distributed to parents and staff to capture usability expectations, preferred reporting formats, and comfort levels with digital systems. This quantitative data will help prioritize system features.
- **Document Review:** Analysis of enrollment records, billing logs, and communication templates will provide baseline data on existing practices. Reviewing these documents will highlight inefficiencies and gaps that the CMS must address.

Together, these methods ensure comprehensive requirements gathering, aligning the system design with both user expectations and institutional needs.

3.3 System Analysis

System analysis involves identifying the core entities, their attributes, and relationships within the childcare environment. According to Sommerville (2016), effective system analysis reduces the risk of failure by ensuring that the system aligns with organizational goals.

The domain objects for the CMS include:

- **Child:** Represents each enrolled child, with attributes such as personal details, enrollment status, and attendance records.
- **Parent:** Represents guardians responsible for children, with attributes including contact information, billing records, and communication history.
- **Staff:** Represents caregivers and administrators, with attributes such as schedules, assignments, and performance logs.
- **Report:** Represents structured outputs in CSV/JSON formats, including attendance summaries, billing statements, and activity logs.
- **User:** Represents all actors interacting with the system (administrators, staff, or parents), with defined roles and permissions to ensure data security and accountability.

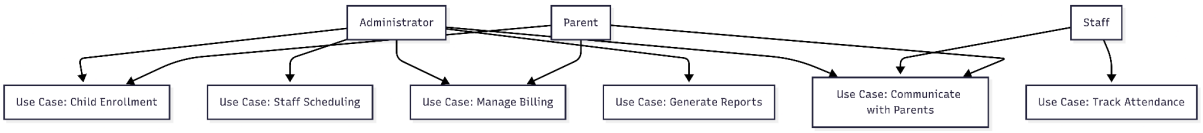
Relationships among these objects will model how administrators manage enrollment, staff record attendance, parents receive updates, and reports are generated for decision-making. This analysis ensures modularity, scalability, and alignment with childcare workflows.

3.4 System Design

UML diagrams will be used to model system functionality:

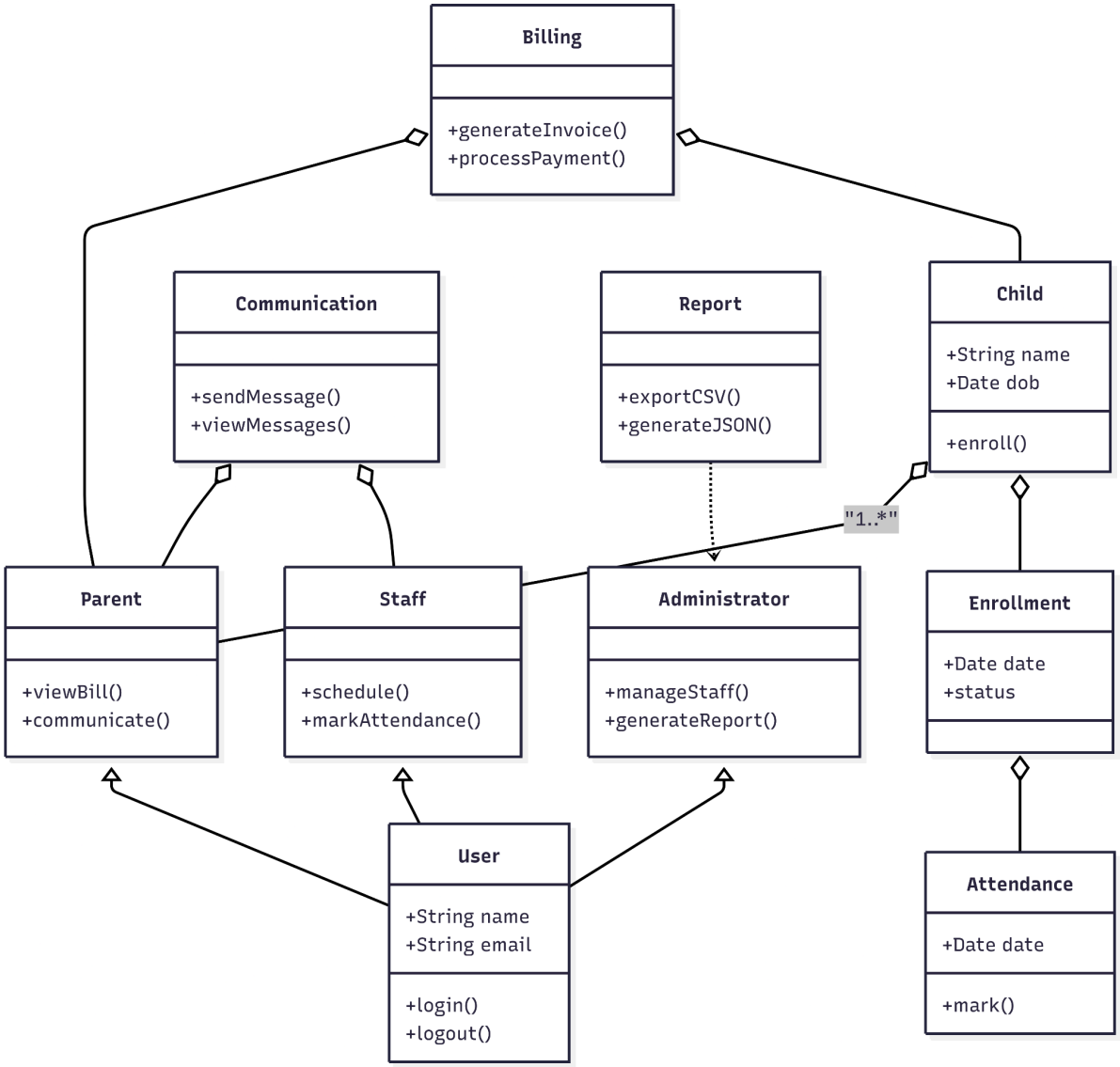
- **Use Case Diagram:** Illustrates actor interactions (Administrator, Parent, Staff).

Figure 3.4.1: Use case Diagram



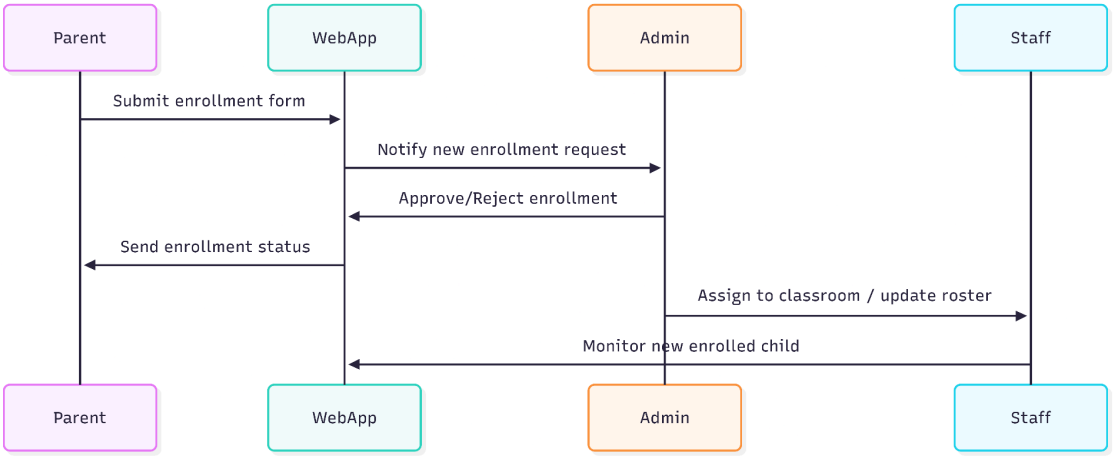
- **Class Diagram:** Defines system modules and their relationships.

Figure 3.4.2: Class Diagram



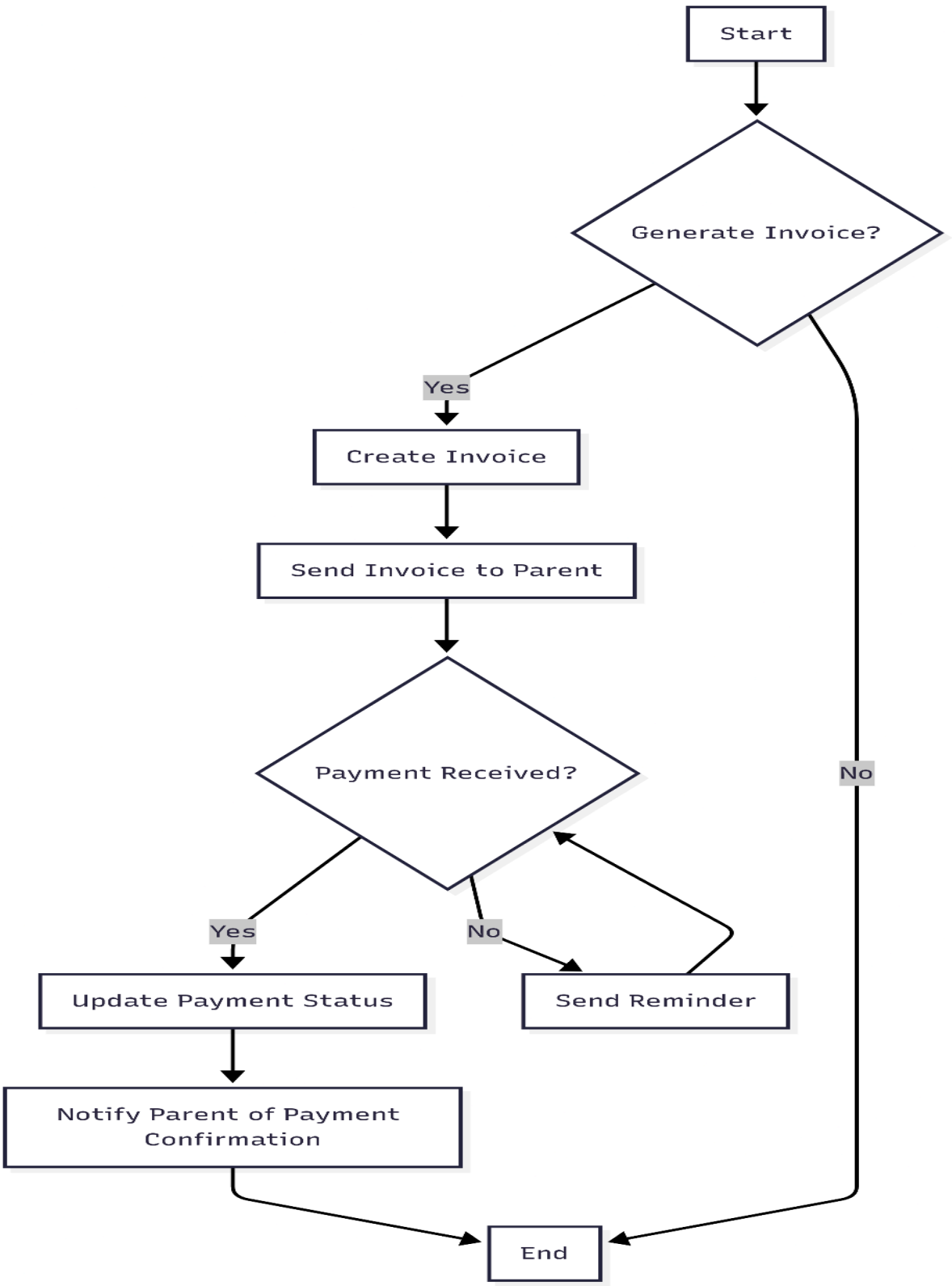
- **Sequence Diagram:** Models workflows such as billing or attendance tracking.

Figure 3.4.3: Sequence Diagram



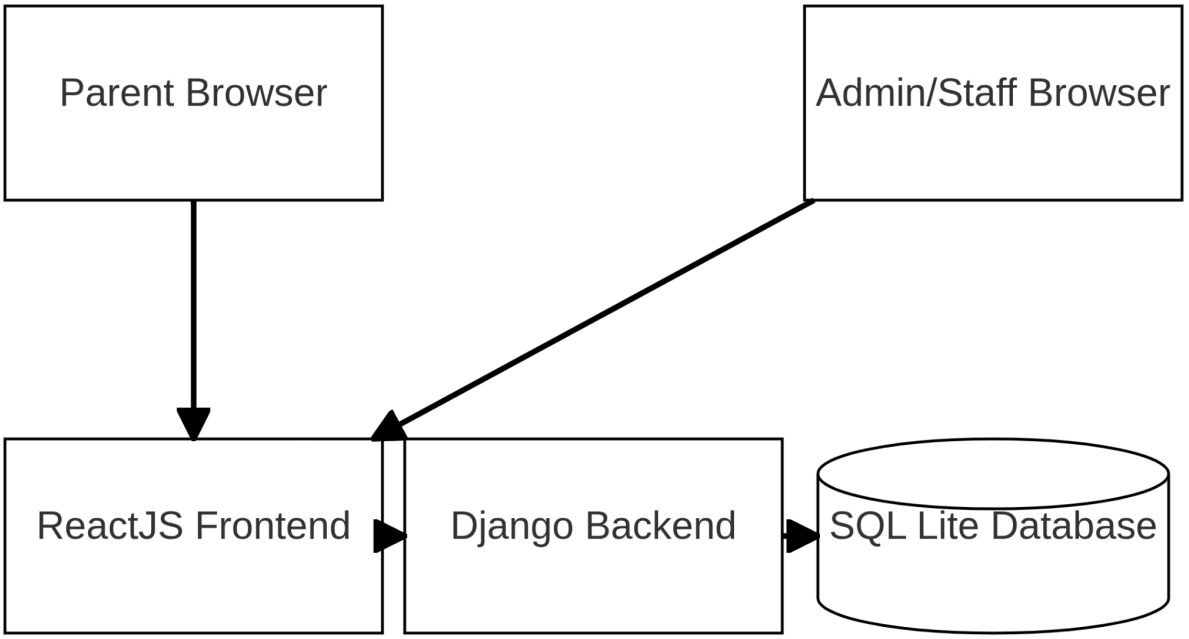
- **Activity Diagram:** Represents processes like enrollment and reporting.

Figure 3.4.4: Activity Diagram



- **Deployment Diagram:** Shows system components (frontend, backend, database).

Figure 3.4.5: Deployment Diagram



3.5 System Development

The Childcare Management System (CMS) will be developed using modern web technologies to ensure scalability, responsiveness, and ease of use. The choice of technologies is guided by their proven reliability, open-source support, and adaptability to local contexts.

- **Frontend:** The user interface will be built using **ReactJS** combined with **Bootstrap CDN**. ReactJS provides a component-based architecture that enhances modularity and reusability, while Bootstrap ensures responsive design across devices, making the system accessible on desktops, tablets, and smartphones (Pressman, 2018).
- **Backend:** The server-side logic will be implemented using **Django**, which offers lightweight, event-driven architecture suitable for real-time applications. Express provides a robust framework for handling HTTP requests, authentication, and API integration.
- **Database:** The system will utilize **MySQL** or **MongoDB** depending on deployment needs. MySQL offers structured relational data management, while MongoDB provides flexibility for unstructured data such as communication logs. Both options ensure scalability and efficient query handling (Sommerville, 2016).
- **Dependency Management: Yarn** will be used to manage project dependencies, ensuring consistent builds and faster installations.

- **Reporting:** The CMS will generate structured reports in **CSV/JSON formats**, enabling administrators to export data for analysis, auditing, and sharing with stakeholders.

Development will follow Agile Scrum methodology, with sprints focusing on individual modules such as enrollment, attendance, billing, and communication. Each sprint will involve requirement refinement, coding, testing, and stakeholder feedback, ensuring iterative improvement and alignment with user needs.

3.6 System Testing

Testing is critical to ensure the reliability, usability, and security of the CMS. Multiple testing strategies will be employed:

- **Functionality Testing:** Verifies that each module performs its intended task, such as accurate enrollment, attendance tracking, billing calculations, and communication delivery.
- **Usability Testing:** Engages parents and staff to assess ease of use, clarity of dashboards, and accessibility of reports. Feedback will be incorporated to improve user experience (UNESCO, 2021).
- **Security Testing:** Validates data protection mechanisms, including encryption, authentication, and role-based access control. This ensures compliance with privacy standards and safeguards sensitive child and parent data.
- **Performance Testing:** Evaluates scalability under multiple concurrent users, ensuring the system can handle peak usage without degradation. Tools such as Selenium and JMeter will be employed for automated testing.

3.7 Ethical Considerations

Ethical considerations are central to the development and deployment of the CMS. The system will:

- Ensure **privacy of children's data**, with strict access controls and encryption.
- Obtain **informed consent** from parents and administrators before data collection or system deployment.

- Be deployed in **controlled environments** during testing to avoid disruption of childcare services.
- Comply with national and institutional policies on data protection and child welfare.

By prioritizing ethics, the CMS will build trust among stakeholders and align with international standards for childcare management (Abu Yazid et al., 2025).

3.8 Limitations of the Methodology

Despite its strengths, the methodology may face certain limitations:

- **Restricted Access to Records:** Some childcare centers may limit access to sensitive data, affecting the comprehensiveness of requirements gathering.
- **Resource Constraints:** Limited budgets may restrict hardware acquisition, internet connectivity, or staff training.
- **Simulated Environments:** Testing may rely on simulated datasets rather than live data, which could affect accuracy in real-world deployment.
- **Resistance to Change:** Administrators and staff accustomed to manual processes may be hesitant to adopt digital systems.

These limitations will be mitigated through stakeholder engagement, phased implementation, and training programs to build user confidence.

3.9 Conclusion

The methodology provides a clear roadmap for developing a robust, user-friendly, and secure Childcare Management System tailored to Mombasa's environment. By combining SDLC with Agile Scrum, the project ensures structured development while remaining flexible to evolving requirements. The integration of modern web technologies, comprehensive testing, and ethical safeguards will deliver a system that enhances efficiency, accountability, and communication in childcare centers.

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APPENDICES

Appendix A: Sample Questionnaire

1. How do you currently manage child enrollment and attendance?
2. What challenges do you face with billing and communication?
3. Which features would be most useful in a childcare management system?
4. How comfortable are you with using digital platforms for childcare management?
5. What improvements would you suggest for a system designed for Mombasa's childcare centers?

Appendix B: Project Time Frame

Phase	**Activities**	**Duration**
Requirement Analysis	Interviews, questionnaires, document reviews	3 weeks
System Design	UML diagrams, architecture definition	1 weeks
Development	Implement core modules	3 weeks
Integration & Reporting	Consolidate modules, reporting features	2 weeks
Testing & Evaluation	Functionality, usability, performance testing	3 weeks
Documentation & Final Report	Prepare manuals and final report	1 week

Appendix C: Project Budget

Item	Description	Estimated Cost (KES)
Software Tools	ReactJS, Yarn, Bootstrap (open-source)	0
Hosting & Domain	Web hosting and domain registration	6,000
Data Collection	Printing questionnaires, interviews	5,000
Documentation	Printing and binding	2,000
Miscellaneous	Contingency	5,000
Total		**18,000**